Homework 1: Running Nextflow Pipelines

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GitHub Repository: https://github.com/Hiroomy-cat/ITMO_Bioinformatics_pipelines_2026

1. Installation of Nextflow

Download and install Nextflow

```
curl -s https://get.nextflow.io | bash
```

```
hiroomy@DESKTOP-EO5KC86:~$ curl -s https://get.nextflow.io | bash

      N E X T F L O W
      version 25.10.4 build 11173
      created 10-02-2026 15:17 UTC (18:17 MSD)
      cite doi:10.1038/nbt.3820
      http://nextflow.io


Nextflow installation completed. Please note:
- the executable file `nextflow` has been created in the folder: /home/hiroomy
- you may complete the installation by moving it to a directory in your $PATH
```

Make it executable and move to a folder in PATH

```
chmod +x nextflow
sudo mv nextflow /usr/local/bin/
```

Verification:

```
nextflow -v
```

```
hiroomy@DESKTOP-EO5KC86:~$ nextflow -v
nextflow version 25.10.4.11173
```

2. Running rnaseq-nf on Test Data

I used the standard tutorial pipeline from nextflow-io/rnaseq-nf. The pipeline was executed using two different package managers/containers to ensure reproducibility.

```
hiroomy@DESKTOP-EO5KC86:~$ nextflow pull nextflow-io/rnaseq-nf
Checking nextflow-io/rnaseq-nf ...
 downloaded from https://github.com/nextflow-io/rnaseq-nf.git - revision: ca20a6dfd2 [master]
```

<u>A. Running with Docker</u>

Docker handles all software dependencies (FastQC, Salmon, MultiQC) automatically within a container.

Command:

```
nextflow run nextflow-io/rnaseq-nf -profile docker
```

```
hiroomy@DESKTOP-EO5KC86:~$ nextflow run nextflow-io/rnaseq-nf -profile docker

 N E X T F L O W  ~  version 25.10.4

Launching `https://github.com/nextflow-io/rnaseq-nf` [maniac_brown] DSL2 - revision: ca20a6dfd2 [master]

R N A S E Q - N F   P I P E L I N E
===================================
transcriptome: /home/hiroomy/.nextflow/assets/nextflow-io/rnaseq-nf/data/ggal/ggal_1_48850000_49020000.Ggal71.500bpflank
.fa
reads        : /home/hiroomy/.nextflow/assets/nextflow-io/rnaseq-nf/data/ggal/ggal_gut_{1,2}.fq
outdir       : results

executor >  local (4)
[87/7564ba] RNASEQ:INDEX (ggal_1_48850000_49020000) | 1 of 1 ✔
[6c/9a7a5a] RNASEQ:FASTQC (ggal_gut)                | 1 of 1 ✔
[52/5a9fba] RNASEQ:QUANT (ggal_gut)                 | 1 of 1 ✔
[af/191cc4] MULTIQC                                 | 1 of 1 ✔

Done! Open the following report in your browser --> results/multiqc_report.html

Completed at: 09-Apr-2026 17:04:25
Duration    : 5m 44s
CPU hours   : 0.2
Succeeded   : 4
```

B. Running with Conda

Conda creates a local environment and downloads the necessary bioinformatics tools specified in the pipeline profile.

Command:

#for nextflow I create new environment

```
conda create -n rnaseq_nf_new python=3.10 -y
conda activate rnaseq_nf_new
conda install -c bioconda nextflow=25.10.4 -y

nextflow run nextflow-io/rnaseq-nf -profile conda --outdir my_results
```

```
(rnaseq_nf_new) hiroomy@DESKTOP-EO5KC86:~$ nextflow run nextflow-io/rnaseq-nf -profile conda --outdir my_results

 N E X T F L O W  ~  version 25.10.4

Launching `https://github.com/nextflow-io/rnaseq-nf` [pedantic_ardinghelli] DSL2 - revision: ca20a6dfd2 [master]

R N A S E Q - N F   P I P E L I N E
===================================
transcriptome: /home/hiroomy/.nextflow/assets/nextflow-io/rnaseq-nf/data/ggal/ggal_1_48850000_49020000.Ggal71.500bpflank
.fa
reads        : /home/hiroomy/.nextflow/assets/nextflow-io/rnaseq-nf/data/ggal/ggal_gut_{1,2}.fq
outdir       : my_results

executor >  local (4)
[70/a9ad4f] RNASEQ:INDEX (ggal_1_48850000_49020000) | 1 of 1 ✔
[ea/569771] RNASEQ:FASTQC (ggal_gut)                | 1 of 1 ✔
[60/bec1c4] RNASEQ:QUANT (ggal_gut)                 | 1 of 1 ✔
[e0/c5d3e9] MULTIQC                                 | 1 of 1 ✔

Done! Open the following report in your browser --> my_results/multiqc_report.html

Completed at: 09-Apr-2026 17:38:36
Duration    : 6m 35s
CPU hours   : (a few seconds)
Succeeded   : 4
```

3. Running the Pipeline on Custom NCBI Data

For this task, I selected a small RNA-seq dataset from the NCBI SRA.

Dataset Details:

SRA Accession: SRR1553606

Organism: *Zaire ebolavirus*

Step 1: Downloading the data

I tried using fastq-dump tool to download the data, but it kept giving me a server error due to my slow internet connection. So I downloaded the data directly.

```
fastq-dump --split-files -X 100000 SRR1553606
```

#and after

```
wget ftp://ftp.sra.ebi.ac.uk/vol1/fastq/SRR155/006/SRR1553606/SRR1553606_1.fastq.gz wget
```

```
SRR1553606_1.fastq.gz         100%[===================================================>]  25.01M   297KB/s    in 2m 29s

2026-04-09 19:02:21 (172 KB/s) - 'SRR1553606_1.fastq.gz' saved [26221960]
```

```
ftp://ftp.sra.ebi.ac.uk/vol1/fastq/SRR155/006/SRR1553606/SRR1553606_2.fastq.gz
```

```
SRR1553606_2.fastq.gz         100%[===================================================>]  25.14M   306KB/s    in 2m 1s

2026-04-09 19:04:47 (212 KB/s) - 'SRR1553606_2.fastq.gz' saved [26363005]
```

Step 2: Running the pipeline

I ran the rnaseq-nf pipeline by pointing to the newly downloaded FASTQ files.

Command:

```
nextflow run nextflow-io/rnaseq-nf \
  --reads 'data/SRR1553606_{1,2}.fastq.gz' \
  -profile conda
```

Technical Challenge & Solution:

When running the pipeline on the yeast dataset (SRR1553606), the process initially failed because the default reference transcriptome in the rnaseq-nf pipeline is for *Gallus gallus* . This caused a 0% alignment rate in Salmon.

I downloaded the correct *Saccharomyces cerevisiae transcriptome* from Ensembl and provided it to the pipeline using the --transcriptome parameter. After this adjustment, the pipeline completed successfully.

I encountered a 'Not a valid path value' error during the indexing stage. To fix this, I unzipped the transcriptome FASTA file and used absolute paths (using the $PWD variable) to ensure Nextflow could correctly locate the input files from its internal working directories.

New command

```
nextflow run nextflow-io/rnaseq-nf \
```

```
  --reads "$PWD/data/SRR1553606_{1,2}.fastq.gz" \
  --transcriptome "$PWD/data/Saccharomyces_cerevisiae.R64-1-1.cdna.all.fa" \
  -profile conda
```

```
N E X T F L O W  ~  version 25.10.4

Launching `https://github.com/nextflow-io/rnaseq-nf` [pedantic_volta] DSL2 - revision: ca20a6dfd2 [master]

R N A S E Q - N F   P I P E L I N E
===================================
transcriptome: /home/hiroomy/hw/data/Saccharomyces_cerevisiae.R64-1-1.cdna.all.fa
reads        : /home/hiroomy/hw/data/SRR1553606_{1,2}.fastq.gz
outdir       : results

[-        ] RNASEQ:INDEX  -
[-        ] RNASEQ:INDEX  | 0 of 1
executor >  local (1)
executor >  local (1)
executor >  local (2)
executor >  local (2)
executor >  local (3)
executor >  local (4)
[8e/325b06] RNASEQ:INDEX (Saccharomyces_cerevisiae) | 1 of 1 ✔
[49/04ff60] RNASEQ:FASTQC (SRR1553606)              | 1 of 1 ✔
[d2/5885dd] RNASEQ:QUANT (SRR1553606)               | 1 of 1 ✔
[e7/ed5edd] MULTIQC                                 | 1 of 1 ✔

Done! Open the following report in your browser --> results/multiqc_report.html

Completed at: 09-Apr-2026 19:25:12
Duration    : 3m 51s
CPU hours   : (a few seconds)
Succeeded   : 4
```

---

4. Results and Deliverables

The pipeline successfully generated the following outputs in the results directory:

FastQC reports: Quality control of the raw reads.

Salmon output: Transcript quantification.

MultiQC report: A consolidated HTML report of the entire run.

```
(rnaseq_nf_new) hiroomy@DESKTOP-EO5KC86:~/hw$ cd results/
(rnaseq_nf_new) hiroomy@DESKTOP-EO5KC86:~/hw/results$ ls
fastqc_SRR1553606_logs  multiqc_report.html
```